

Lecture Note 5: Intro to Estimation, Sufficient Statistics and Rao-Blackwell

14.380: Statistical Methods in Economics, Fall 2026

1 Estimators and Properties of Estimators

Let $X = (X_1, \dots, X_n)'$ be a vector of observations drawn from some parametric distribution $F(x; \theta)$. Let $\hat{\theta} = T(X)$ denote an *estimator* of θ , and a realization of an estimator is referred to as an *estimate*.

Example 1. If $X_1, \dots, X_n$ is a random sample from some distribution with mean μ and variance σ^2 , then sample average $\hat{\mu} = \bar{X}_n$ is an estimator of the population mean, and sample variance $\hat{\sigma}^2 = s^2 = \sum_{i=1}^n (X_i - \bar{X}_n)^2 / (n-1)$ is an estimator of the population variance.

1.1 Unbiasedness

Let X be our data. Let $\hat{\theta} = T(X)$ be an estimator where T is some function. We say that $\hat{\theta}$ is *unbiased* for θ if $\mathbb{E}_\theta[T(X)] = \theta$ for all possible values of θ , where $\mathbb{E}_\theta$ denotes the expectation when θ is the true parameter value. The *bias* of $\hat{\theta}$ is defined by

$$\text{Bias}(\hat{\theta}) = \mathbb{E}_\theta[\hat{\theta}] - \theta.$$

Thus, the concept of unbiasedness means that we are on average correct. For example, if X is a random sample $X_1, \dots, X_n$ from some distribution with mean μ and variance σ^2 , then, as we have already seen, $\mathbb{E}[\hat{\mu}] = \mu$ and $\mathbb{E}[s^2] = \sigma^2$. Thus, sample average and sample variance are unbiased estimators of population mean and population variance correspondingly.

There are some cases when unbiased estimators do not exist.

Example 2. Let $X_1, \dots, X_n$ be a random sample from a Bernoulli(p) distribution. Suppose that our parameter of interest is $\theta = 1/p$. Let $\hat{\theta} = T(X)$ be some estimator. Then

$$\mathbb{E}[\hat{\theta}] = \sum_{(x_1, \dots, x_n) \in \{0,1\}^n} T(x_1, \dots, x_n) P\{(X_1, \dots, X_n) = (x_1, \dots, x_n)\}.$$

We know that for any $(x_1, \dots, x_n) \in \{0,1\}^n$,

$$P\{(X_1, \dots, X_n) = (x_1, \dots, x_n)\} = p^{\sum x_i} (1-p)^{\sum (1-x_i)}$$

which is a polynomial of degree n in p . Therefore, $\mathbb{E}[\hat{\theta}]$ is a polynomial of degree at most n in p . However, $1/p$ is not a polynomial at all. Hence, there are no unbiased estimators in this case.

The example above is very typical in the sense that parameter p has an unbiased estimator $\hat{p} = \frac{1}{n} \sum_{i=1}^n X_i$, but the parameter of interest is a non-linear function of p . Notice that $\mathbb{E}\left[\frac{1}{\xi}\right] \neq \frac{1}{\mathbb{E}[\xi]}$, and the bias appears from the non-linear transformation. This bias can be partially corrected by bootstrap as we discuss next.

Bootstrap bias correction: Another task for which the bootstrap is used is bias-correction. Suppose $\mathbb{E}[Z] = \mu$ and we are interested in a non-linear function of μ , say $\theta = g(\mu)$. Here Z may be a random variable coming from transformations of the observed data: $Z_i = h(X_i)$. We do have an unbiased estimate of μ , say $\hat{\mu} = \bar{Z} = \frac{1}{n} \sum_{i=1}^n Z_i$. We may try to use this in order to estimate θ : $\hat{\theta} = g(\bar{Z})$.

This estimator $\hat{\theta}$ is reasonable but it is biased unless $g(\cdot)$ is linear. The bias is $\text{Bias} = \mathbb{E}[\hat{\theta}] - g(\mu)$. We can estimate the bias using the bootstrap:

1. For each $b = 1, \dots, B$ generate a bootstrap sample, $\{Z_{ib}^*\}$ from the set $\{Z_1, \dots, Z_n\}$ with replacement;
2. Calculate $\bar{Z}_b^* = \frac{1}{n} \sum_{i=1}^n Z_{ib}^*$;
3. Estimate $\theta_b^* = g(\bar{Z}_b^*)$;
4. $\text{Bias}^* = \frac{1}{B} \sum_{b=1}^B \theta_b^* - \hat{\theta} \approx \text{Bias}$.
5. Use $\tilde{\theta} = \hat{\theta} - \text{Bias}^*$ as your estimate.

Why does it work? Let's denote $G_1(\mu) = \frac{dg(\mu)}{d\mu}$ and $G_2(\mu) = \frac{d^2g(\mu)}{d\mu^2}$. Notice that if CLT works we have $\sqrt{n}(\bar{Z} - \mu) \xrightarrow{d} N(0, \sigma_z^2)$, where $\sigma_z^2 = \text{Var}(Z_i)$; or $\bar{Z} - \mu = O_p(1/\sqrt{n})$. Then

$$\hat{\theta} - \theta = g(\bar{Z}) - g(\mu) = G_1(\mu)(\bar{Z} - \mu) + \frac{1}{2}G_2(\mu)(\bar{Z} - \mu)^2 + o_p\left(\frac{1}{n}\right),$$

$$\text{Bias} = \mathbb{E}(\hat{\theta} - \theta) = \frac{1}{2}G_2(\mu)\mathbb{E}(\bar{Z} - \mu)^2 = \frac{1}{2}G_2(\mu)\frac{\sigma_z^2}{n} + o\left(\frac{1}{n}\right),$$

and similarly

$$\text{Bias}^* = \frac{1}{2}G_2(\bar{Z})\frac{s_z^2}{n} + o_p\left(\frac{1}{n}\right).$$

As a result,

$$\text{Bias}^* - \text{Bias} = o_p\left(\frac{1}{n}\right).$$

This procedure eliminates the leading term in bias ($O(1/n)$), but not the whole of the bias. The remaining bias is of order $o(1/n)$. Notice that in principle there was an asymptotic approach to eliminate bias as well (as we did get the formula for the leading term $\frac{1}{2}G_2(\mu)\frac{\sigma_z^2}{n} + o(\frac{1}{n})$). One in principle could have approximated it by $\frac{1}{2}G_2(\bar{Z})\frac{s_z^2}{n}$, but the bootstrap does this automatically.

Example 3. Assume we wish to estimate the skewness of a distribution $\theta = \frac{\mathbb{E}[(X - \mathbb{E}[X])^3]}{[\text{Var}(X)]^{3/2}}$. A natural estimate is

$$\hat{\theta} = \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^3}{s^3}.$$

This is a non-linear function of $\bar{Z} = (\bar{X}, \frac{1}{n} \sum_{i=1}^n X_i^2, \frac{1}{n} \sum_{i=1}^n X_i^3)$. As such it will most likely have bias, which we can correct with the bootstrap.

Example 4. Unbiasedness is not limited to i.i.d. sampling; it is also the first property one checks for estimators of causal effects. Consider a finite population of N units, as in Lecture Note 2, Section 4, where each unit i now carries *two* fixed numbers: an outcome $y_i(1)$ that would be realized if unit i were treated, and an outcome $y_i(0)$ that would be realized if it were not. The estimand is the *average treatment effect*

$$\tau = \frac{1}{N} \sum_{i=1}^N (y_i(1) - y_i(0)).$$

In a *completely randomized experiment*, the researcher selects n_1 units uniformly at random to receive treatment ($D_i = 1$) and the remaining $n_0 = N - n_1$ serve as controls ($D_i = 0$); the observed outcome is $Y_i = y_i(D_i)$. The natural estimator is the difference in means,

$$\hat{\tau} = \frac{1}{n_1} \sum_{i: D_i=1} Y_i - \frac{1}{n_0} \sum_{i: D_i=0} Y_i.$$

Is $\hat{\tau}$ unbiased for τ ? Yes — and the proof is a calculation we have already done. The treated group is a simple random sample of size n_1 from the finite population, so its sample mean is an unbiased estimator of the population mean of the $y_i(1)$ by the theorem in Lecture Note 2, Section 4; by the same argument, the control mean is unbiased for the population mean of the $y_i(0)$. Taking differences,

$$\mathbb{E}[\hat{\tau}] = \frac{1}{N} \sum_{i=1}^N y_i(1) - \frac{1}{N} \sum_{i=1}^N y_i(0) = \tau.$$

As in Lecture Note 2, Section 4, the potential outcomes are fixed numbers and the only randomness is the assignment chosen by the researcher, so unbiasedness here is a statement about the distribution of treatment assignments. This calculation is due to Neyman (1923). Characterizing $V(\hat{\tau})$ is more subtle — the same random assignment determines both group means, so they are not independent; see Imbens and Rubin (2015) for a complete treatment.

1.2 Mean Square Error

Another concept that is used to evaluate the performance of estimators is the *mean squared error* (MSE). By definition,

$$\text{MSE}(\hat{\theta}) = \mathbb{E}_{\theta}[(\hat{\theta} - \theta)^2].$$

The theorem below gives a useful decomposition for MSE:

Theorem 1. $\text{MSE}(\hat{\theta}) = \text{Bias}^2(\hat{\theta}) + V(\hat{\theta})$.

Proof.

$$\begin{aligned}
\mathbb{E}[(\hat{\theta} - \theta)^2] &= \mathbb{E}[(\hat{\theta} - \mathbb{E}[\hat{\theta}] + \mathbb{E}[\hat{\theta}] - \theta)^2] \\
&= \mathbb{E}[(\hat{\theta} - \mathbb{E}[\hat{\theta}])^2 + (\mathbb{E}[\hat{\theta}] - \theta)^2 + 2(\hat{\theta} - \mathbb{E}[\hat{\theta}])(\mathbb{E}[\hat{\theta}] - \theta)] \\
&= V(\hat{\theta}) + \text{Bias}^2(\hat{\theta}) + 2\mathbb{E}[\hat{\theta} - \mathbb{E}[\hat{\theta}]](\mathbb{E}[\hat{\theta}] - \theta) \\
&= V(\hat{\theta}) + \text{Bias}^2(\hat{\theta}).
\end{aligned}$$

□

We often compare estimators on the basis of their MSE. In the next lecture, we will introduce other criteria that are used to compare estimators.

Quite often there is a trade-off between the bias of the estimator and its variance. Thus, we may prefer a slightly biased estimator to an unbiased one if the former has much smaller variance in comparison to the latter one.

Example 5. Let $X_1, \dots, X_n$ be a random sample from $N(\mu, \sigma^2)$. Let $\hat{\sigma}_1^2 = s^2 = \sum_{i=1}^n (X_i - \bar{X}_n)^2 / (n-1)$ and $\hat{\sigma}_2^2 = \sum_{i=1}^n (X_i - \bar{X}_n)^2 / n$ be two estimators of σ^2 . We know that $\mathbb{E}[\hat{\sigma}_1^2] = \sigma^2$. So $\mathbb{E}[\hat{\sigma}_2^2] = ((n-1)/n)\mathbb{E}[\hat{\sigma}_1^2] = ((n-1)/n)\sigma^2$, and $\text{Bias}(\hat{\sigma}_2^2) = -\sigma^2/n$. We also know that $(n-1)\hat{\sigma}_1^2/\sigma^2 \sim \chi^2(n-1)$.

What is $V(\chi^2(n-1))$? Let $\xi_1, \dots, \xi_{n-1}$ be a random sample from $N(0, 1)$. Then $\xi = \xi_1^2 + \dots + \xi_{n-1}^2 \sim \chi^2(n-1)$. By linearity of expectation, $\mathbb{E}[\xi] = (n-1)$. By independence,

$$\begin{aligned}
\mathbb{E}[\xi^2] &= \mathbb{E}[(\xi_1^2 + \dots + \xi_{n-1}^2)^2] \\
&= \sum_{i=1}^{n-1} \mathbb{E}[\xi_i^4] + 2 \sum_{1 \leq i < j \leq n-1} \mathbb{E}[\xi_i^2 \xi_j^2] \\
&= 3(n-1) + 2 \sum_{1 \leq i < j \leq n-1} \mathbb{E}[\xi_i^2] \mathbb{E}[\xi_j^2] \\
&= 3(n-1) + (n-1)(n-2) \\
&= (n-1)(n+1),
\end{aligned}$$

since $\mathbb{E}[\xi_i^4] = 3$. So

$$V(\xi) = \mathbb{E}[\xi^2] - (\mathbb{E}[\xi])^2 = (n-1)(n+1) - (n-1)^2 = 2(n-1).$$

Thus, $V(\hat{\sigma}_1^2) = V(\sigma^2 \xi / (n-1)) = 2\sigma^4 / (n-1)$ and $V(\hat{\sigma}_2^2) = ((n-1)/n)^2 V(\hat{\sigma}_1^2) = 2\sigma^4(n-1)/n^2$. Finally, $\text{MSE}(\hat{\sigma}_1^2) = 2\sigma^4/(n-1)$ and

$$\text{MSE}(\hat{\sigma}_2^2) = \sigma^4/n^2 + 2\sigma^4(n-1)/n^2 = (2n-1)\sigma^4/n^2.$$

So, $\text{MSE}(\hat{\sigma}_1^2) < \text{MSE}(\hat{\sigma}_2^2)$ if and only if $2/(n-1) < (2n-1)/n^2$, which is equivalent to $3n < 1$. So, for any $n \geq 1$, $\text{MSE}(\hat{\sigma}_1^2) > \text{MSE}(\hat{\sigma}_2^2)$ in spite of the fact that $\hat{\sigma}_1^2$ is unbiased.

In general, the idea of minimizing MSE is typically not in agreement with unbiasedness: one may improve MSE if we allow for some bias. Indeed, this intuition lies at the core of the bias-variance tradeoff.

Indeed, we can see this clearly in the following setting. Assume that the parameter set Θ is bounded and $\hat{\theta} = T(X)$ is an unbiased estimator of θ : $\mathbb{E}[T(X)] = \theta$. Take any fixed point $\theta^* \in \Theta$ and shrink the initial estimator towards it:

$$\hat{\theta}_1 = (1 - c)T(X) + c\theta^*.$$

Here c characterizes the amount of shrinkage. The new estimator is somewhat biased $\text{Bias}(\hat{\theta}_1) = c(\theta^* - \theta)$ but is less dispersed $\text{Var}(\hat{\theta}_1) = (1 - c)^2 \text{Var}(\hat{\theta})$. So, we have

$$\text{MSE}(\hat{\theta}_1) = c^2(\theta^* - \theta)^2 + (1 - c)^2 \text{Var}(\hat{\theta}).$$

One may calculate the derivative of MSE with respect to c at $c = 0$ and find that it is negative, and thus some small positive amount of shrinkage $c > 0$ will improve the efficiency of the initial estimator.

The next example works out this bias-variance tradeoff fully in a simple class of estimators: weighted averages of the data.

Example 6. Let $X_1, \dots, X_n$ be a random sample from some distribution with mean μ and variance σ^2 (both unknown). Consider estimating μ by a weighted average

$$\hat{\mu} = \sum_{i=1}^n \omega_i X_i,$$

where $\omega_1, \dots, \omega_n$ are nonrandom weights.

First, when is $\hat{\mu}$ unbiased? We have $\mathbb{E}[\hat{\mu}] = \sum_{i=1}^n \omega_i \mathbb{E}[X_i] = \mu \sum_{i=1}^n \omega_i$, so $\hat{\mu}$ is unbiased if and only if $\sum_{i=1}^n \omega_i = 1$.

Second, among all unbiased estimators of this form, which has the smallest variance? By independence, $V(\hat{\mu}) = \sigma^2 \sum_{i=1}^n \omega_i^2$, so we need to minimize $\sum_{i=1}^n \omega_i^2$ subject to $\sum_{i=1}^n \omega_i = 1$. The Lagrangian is $\sum_{i=1}^n \omega_i^2 - \lambda(\sum_{i=1}^n \omega_i - 1)$, and the first-order condition for each ω_i is $2\omega_i = \lambda$. Thus, all weights are equal, and the constraint forces $\omega_i = 1/n$. The sample average $\bar{X}_n$ is therefore the minimum-variance estimator among unbiased weighted averages.

Third, suppose we drop unbiasedness and instead minimize the MSE. By the bias-variance decomposition,

$$\text{MSE}(\hat{\mu}) = \left[\mu \left(\sum_{i=1}^n \omega_i - 1 \right) \right]^2 + \sigma^2 \sum_{i=1}^n \omega_i^2.$$

This objective is strictly convex in the weights, so the first-order conditions are necessary and sufficient. Differentiating with respect to ω_i gives $2\mu^2(\sum_{j=1}^n \omega_j - 1) + 2\sigma^2\omega_i = 0$. The first term is the same for every i , so again all weights are equal, say $\omega_i = \omega$. Substituting, $\mu^2(n\omega - 1) + \sigma^2\omega = 0$,

which gives

$$\omega_i = \frac{\mu^2}{\sigma^2 + n\mu^2}.$$

The resulting estimator is biased: $\text{Bias}(\hat{\mu}) = \mu(n\omega - 1) = -\frac{\mu\sigma^2}{\sigma^2 + n\mu^2}$, which is nonzero whenever $\mu \neq 0$. Note that the MSE-optimal weights can be written as $\omega_i = (1-c)/n$ with $c = \sigma^2/(\sigma^2 + n\mu^2) \in (0, 1)$; that is, the MSE-optimal estimator shrinks the sample average $\bar{X}_n$ towards $\theta^* = 0$, exactly as in the shrinkage argument above. Of course, the optimal weights depend on the unknown (μ, σ^2) , so this estimator is infeasible in practice; the value of the example is that it makes the bias-variance tradeoff fully explicit.

1.3 Asymptotic Properties

1.3.1 Consistency

Imagine a thought experiment in which the number of observations n increases without bound, i.e. $n \rightarrow \infty$. Suppose that for each n , we have an estimator $\hat{\theta}_n$.

We say that $\hat{\theta}_n$ is *consistent* for θ if $\hat{\theta}_n \rightarrow_p \theta$.

Example 7. Let $X_1, \dots, X_n$ be a random sample from some distribution with mean μ and variance σ^2 . Let $\hat{\mu} = \hat{\mu}_n = \bar{X}_n$ be our estimator of μ and $s^2 = s_n^2 = \sum_{i=1}^n (X_i - \bar{X}_n)^2 / (n-1)$ be our estimator of σ^2 . By the Law of large numbers, we know that $\hat{\mu} \rightarrow_p \mu$ as $n \rightarrow \infty$. In addition,

$$\begin{aligned} s^2 &= \sum_{i=1}^n (X_i - \bar{X}_n)^2 / (n-1) \\ &= \sum_{i=1}^n (X_i - \mu)^2 / (n-1) - (n/(n-1))(\bar{X}_n - \mu)^2 \\ &= (n/(n-1)) \left(\sum_{i=1}^n (X_i - \mu)^2 / n \right) - (n/(n-1))(\bar{X}_n - \mu)^2 \end{aligned}$$

By the Law of Large Numbers, $\sum_{i=1}^n (X_i - \mu)^2 / n \rightarrow_p \mathbb{E}[(X_i - \mu)^2] = \sigma^2$ and $\bar{X}_n - \mu = \sum_{i=1}^n (X_i - \mu) / n \rightarrow_p \mathbb{E}[X_i - \mu] = 0$. By using the Continuous Mapping Theorem, $(\bar{X}_n - \mu)^2 \rightarrow_p 0$. In addition, $n/(n-1) \rightarrow_p 1$. So, by the Slutsky theorem, $s^2 \rightarrow_p \sigma^2$. So $\hat{\mu}$ and s^2 are consistent for μ and σ^2 correspondingly.

1.3.2 Asymptotic Normality

We say that $\hat{\theta}$ is *asymptotically normal* if there are sequences $\{a_n\}_{n=1}^\infty$ and $\{r_n\}_{n=1}^\infty$ and constant σ^2 such that

$$r_n(\hat{\theta} - a_n) \xrightarrow{d} N(0, \sigma^2).$$

Then r_n is called the *rate of convergence*, a_n is the *asymptotic mean*, and σ^2 is the *asymptotic variance*.

This is particularly useful since we can use the following approximation $\hat{\theta} \stackrel{a}{\sim} N(a_n, \sigma^2/r_n^2)$ when computing probabilities involving $\hat{\theta}$.

In many cases, one can choose $a_n = \theta$ and $r_n = \sqrt{n}$. We will use the concept of asymptotic normality for confidence set construction later on.

Example 8. Let $X_1, \dots, X_n$ be a random sample from some distribution with mean μ and variance σ^2 . Let $\hat{\mu}$ and s^2 be the sample mean and the sample variance correspondingly. Then, by the Central limit theorem, $\sqrt{n}(\hat{\mu} - \mu) \xrightarrow{d} N(0, \sigma^2)$.

As for s^2 ,

$$\sqrt{n}(s^2 - \sigma^2) = (n/(n-1)) \left[\sum_{i=1}^n ((X_i - \mu)^2 - \sigma^2)/\sqrt{n} - (\sqrt{n}(\bar{X}_n - \mu)/n^{1/4})^2 \right] + (\sqrt{n}/(n-1))\sigma^2$$

By the Central limit theorem, $\sum_{i=1}^n ((X_i - \mu)^2 - \sigma^2)/\sqrt{n} \xrightarrow{d} N(0, \tau^2)$ with $\tau^2 = \mathbb{E}[((X_i - \mu)^2 - \sigma^2)^2]$. Note that $\tau^2 = \mu_4 - 2\sigma^2\mathbb{E}[(X_i - \mu)^2] + \sigma^4 = \mu_4 - \sigma^4$ with $\mu_4 = \mathbb{E}[(X_i - \mu)^4]$. By Slutsky theorem, $\sqrt{n}(\bar{X}_n - \mu)/n^{1/4} \rightarrow_p 0$. In addition, $(\sqrt{n}/(n-1))\sigma^2 \rightarrow_p 0$. So, by the Slutsky theorem again, $\sqrt{n}(s^2 - \sigma^2) \xrightarrow{d} N(0, \tau^2)$.

2 Sufficient Statistics

One way to think about constructing estimators is through the lens of “data reduction” – we would like to summarize the information in a dataset with only a small number of statistics. In doing so, a useful concept is a *sufficient statistic*. Intuitively, if θ is an unknown parameter of the probability density of $X = (X_1, \dots, X_n)'$, then a statistic $T(X)$ is sufficient for θ if it summarizes all information in X for θ ; in other words, if you only cared to learn about θ , then you could report $T(X)$ and disregard the rest of the data.

More formally, let $f(x|\theta)$ with $\theta \in \Theta$ be some parametric family. Let $X = (X_1, \dots, X_n)$ be a random sample from distribution $f(x|\theta)$. Suppose we would like to learn parameter value θ from our sample.

Definition 1. Statistic $T(X)$ is sufficient for θ if the conditional distribution of X given $T(X)$ does not depend on θ .

Let $T(X)$ be a sufficient statistic. Consider the pair $(X, T(X))$. Obviously, $(X, T(X))$ contains the same information about θ as X alone, since $T(X)$ is a function of X . But if we know $T(X)$, then X itself has no value for us since its conditional distribution given $T(X)$ is independent of θ . Thus, by observing X (in addition to $T(X)$), we cannot say whether one particular value of parameter θ is more likely than another. Therefore, once we know $T(X)$, we can discard X completely.

Example 9. Suppose Y_1, Y_2 are i.i.d. Bernoulli random variables with $P(Y_i = 1) = \theta$. Let $S = Y_1 + Y_2$. Then, $S \in \{0, 1, 2\}$. Notice that: (i) if $S = 0$, then $Y_1 = Y_2 = 0$ and so $P(Y = (0, 0) | S = 0) = 1$; (ii) if $S = 2$, then $Y_1 = Y_2 = 1$ and so $P(Y = (1, 1) | S = 2) = 1$; (iii) if $S = 1$, then either $Y_1 = 1, Y_2 = 0$ or $Y_1 = 0, Y_2 = 1$ and so $P(Y = (0, 1) | S = 1) = P(Y = (1, 0) | S = 1) = 1/2$. Therefore, we see that $P(Y | S)$ does not depend on θ , and so S is a sufficient statistic.

Example 10. Let $X = (X_1, \dots, X_n)$ be a random sample from $N(\mu, \sigma^2)$. Suppose that σ^2 is known. Thus, the only parameter is μ ($\theta = \mu$). We have already seen that $T(X) = \bar{X}_n \sim N(\mu, \sigma^2/n)$. Let us calculate the conditional distribution of X given $T(X) = t$. First, note that

$$\begin{aligned} \sum_{i=1}^n (x_i - \mu)^2 - n(\bar{x}_n - \mu)^2 &= \sum_{i=1}^n (x_i - \bar{x}_n + \bar{x}_n - \mu)^2 - n(\bar{x}_n - \mu)^2 \\ &= \sum_{i=1}^n (x_i - \bar{x}_n)^2 + 2 \sum_{i=1}^n (x_i - \bar{x}_n)(\bar{x}_n - \mu) \\ &= \sum_{i=1}^n (x_i - \bar{x}_n)^2. \end{aligned}$$

Therefore,

$$\begin{aligned} f_{X|T(X)}(x|T(X) = T(x)) &= \frac{f_X(x)}{f_T(T(x))} \\ &= \frac{\exp\{-\sum_{i=1}^n (x_i - \mu)^2/(2\sigma^2)\}/((2\pi)^{n/2}\sigma^n)}{\exp\{-n(\bar{x}_n - \mu)^2/(2\sigma^2)\}/((2\pi)^{1/2}\sigma/n^{1/2})} \\ &= \exp\{-\sum_{i=1}^n (x_i - \bar{x}_n)^2/(2\sigma^2)\}/((2\pi)^{(n-1)/2}\sigma^{n-1}n^{1/2}), \end{aligned}$$

which is independent of μ . We conclude that $T(X) = \bar{X}_n$ is a sufficient statistic for our parametric family. Note, however, that $\bar{X}_n$ is not sufficient if σ^2 is not known.

2.1 Factorization Theorem

The Factorization Theorem gives a general approach for how to find a sufficient statistic:

Theorem 2 (Factorization Theorem). *Let $f(x|\theta)$ be the pdf of X . Then $T(X)$ is a sufficient statistic if and only if there exist functions $g(t|\theta)$ and $h(x)$ such that*

$$f(x|\theta) = g(T(x)|\theta)h(x).$$

Proof. Let $l(t|\theta)$ be the pdf of $T(X)$. Suppose $T(X)$ is a sufficient statistic. Then $f_{X|T(X)}(x|T(X) = T(x)) = f_X(x|\theta)/l(T(x)|\theta)$ does not depend on θ . Denote it by $h(x)$. Then $f(x|\theta) = l(T(x)|\theta)h(x)$. Denoting l by g yields the result in one direction.

In the other direction, we will give a “sloppy” proof. Denote $A(x) = \{y : T(y) = T(x)\}$. Then

$$l(T(x)|\theta) = \int_{A(x)} f(y|\theta)dy = \int_{A(x)} g(T(y)|\theta)h(y)dy = g(T(x)|\theta) \int_{A(x)} h(y)dy.$$

So

$$\begin{aligned}
f_{X|T(X)}(x|T(X) = T(x)) &= \frac{f(x|\theta)}{l(T(x)|\theta)} \\
&= \frac{g(T(x)|\theta)h(x)}{g(T(x)|\theta) \int_{A(x)} h(y)dy} \\
&= \frac{h(x)}{\int_{A(x)} h(y)dy},
\end{aligned}$$

which is independent of θ . We conclude that $T(X)$ is a sufficient statistic. $\square$

Let us show how to use the factorization in practice through three examples.

Example 11. Let $X_1, \dots, X_n$ be a random sample from $N(\mu, \sigma^2)$ where both μ and σ^2 are unknown, i.e. $\theta = (\mu, \sigma^2)$. Then,

$$\begin{aligned}
f(x|\theta) &= \exp\left\{-\sum_{i=1}^n (x_i - \mu)^2 / (2\sigma^2)\right\} / ((2\pi)^{n/2} \sigma^n) \\
&= \exp\left\{-\left[\sum_{i=1}^n x_i^2 - 2\mu \sum_{i=1}^n x_i + n\mu^2\right] / (2\sigma^2)\right\} / ((2\pi)^{n/2} \sigma^n).
\end{aligned}$$

Thus, $T(X) = (\sum_{i=1}^n X_i^2, \sum_{i=1}^n X_i)$ is a sufficient statistic (here $h(x) = 1$ and g is the whole thing). Note that in this example we actually have a pair of sufficient statistics. In addition, as we have seen before,

$$\begin{aligned}
f(x|\theta) &= \exp\left\{-\left[\sum_{i=1}^n (x_i - \bar{x}_n)^2 + n(\bar{x}_n - \mu)^2\right] / (2\sigma^2)\right\} / ((2\pi)^{n/2} \sigma^n) \\
&= \exp\left\{-[(n-1)s_n^2 + n(\bar{x}_n - \mu)^2] / (2\sigma^2)\right\} / ((2\pi)^{n/2} \sigma^n).
\end{aligned}$$

Thus, $T(X) = (\bar{X}_n, s_n^2)$ is another sufficient statistic. Yet another sufficient statistic is $T(X) = (X_1, \dots, X_n)$. Note that $\bar{X}_n$ is not sufficient in this example.

Example 12. The factorization theorem also applies beyond random samples. Suppose that the random variables $Y_1, \dots, Y_n$ satisfy

$$Y_i = \beta x_i + e_i, \quad i = 1, \dots, n,$$

where $x_1, \dots, x_n$ are fixed constants and $e_1, \dots, e_n$ are i.i.d. $N(0, \sigma^2)$ with σ^2 unknown, i.e. $\theta = (\beta, \sigma^2)$. The Y_i 's are independent but not identically distributed, since each has mean βx_i . The

joint density is

$$\begin{aligned} f(y|\theta) &= \exp\left\{-\sum_{i=1}^n (y_i - \beta x_i)^2 / (2\sigma^2)\right\} / ((2\pi)^{n/2} \sigma^n) \\ &= \exp\left\{-\left[\sum_{i=1}^n y_i^2 - 2\beta \sum_{i=1}^n x_i y_i + \beta^2 \sum_{i=1}^n x_i^2\right] / (2\sigma^2)\right\} / ((2\pi)^{n/2} \sigma^n). \end{aligned}$$

Since the x_i 's are known constants, $\sum_{i=1}^n x_i^2$ is just a number. Thus, the joint density depends on the data only through $(\sum_{i=1}^n y_i^2, \sum_{i=1}^n x_i y_i)$, and by the factorization theorem (with $h(y) = 1$), $T(Y) = (\sum_{i=1}^n Y_i^2, \sum_{i=1}^n x_i Y_i)$ is a two-dimensional sufficient statistic for (β, σ^2) . This is our first example of a sufficient statistic in a regression model.

Example 13. Let $X_1, \dots, X_n$ be a random sample from $U[\theta, 1 + \theta]$. Then $f(x|\theta) = 1$ if $\theta \leq \min_i X_i \leq \max_i X_i \leq 1 + \theta$ and 0 otherwise. In other words, $f(x|\theta) = I\{\theta \leq X_{(1)}\} I\{1 + \theta \geq X_{(n)}\}$. So $T(X) = (X_{(1)}, X_{(n)})$ is sufficient.

2.2 Exponential Families

The examples above share a common structure: the likelihood factors so that the data enter only through a low-dimensional statistic. There is a large class of distributions for which this happens automatically.

Definition 2. A family of distributions $\{f(x; \theta) : \theta \in \Theta\}$ is a **k -parameter exponential family** if its pdf or pmf can be written as

$$f(x; \theta) = h(x) c(\theta) \exp \left(\sum_{j=1}^k w_j(\theta) t_j(x) \right),$$

where $h(\cdot)$ and $t_1(\cdot), \dots, t_k(\cdot)$ are functions of the data alone and $c(\cdot)$ and $w_1(\cdot), \dots, w_k(\cdot)$ are functions of the parameter alone.

Many of the named distributions from Lecture Note 1 take this form. For the Bernoulli distribution,

$$f(x; p) = p^x (1 - p)^{1-x} = (1 - p) \exp \left(x \ln \frac{p}{1 - p} \right),$$

so $h(x) = 1$, $c(p) = 1 - p$, $w(p) = \ln \frac{p}{1-p}$, and $t(x) = x$. For the Poisson distribution,

$$f(x; \lambda) = \frac{e^{-\lambda} \lambda^x}{x!} = \frac{1}{x!} e^{-\lambda} \exp(x \ln \lambda),$$

so $h(x) = 1/x!$, $c(\lambda) = e^{-\lambda}$, $w(\lambda) = \ln \lambda$, and $t(x) = x$. The normal distribution $N(\mu, \sigma^2)$ is a two-parameter exponential family with $t_1(x) = x$ and $t_2(x) = x^2$ (check this as an exercise).

The payoff is that the sufficient statistic for an i.i.d. sample can be read off the family directly.

Theorem 3. Suppose $X_1, \dots, X_n$ are i.i.d. from a k -parameter exponential family. Then,

$$T(\mathcal{X}) = \left(\sum_{i=1}^n t_1(X_i), \dots, \sum_{i=1}^n t_k(X_i) \right)$$

is a sufficient statistic for θ .

Proof. The joint density is

$$\prod_{i=1}^n f(X_i; \theta) = \left(\prod_{i=1}^n h(X_i) \right) c(\theta)^n \exp \left(\sum_{j=1}^k w_j(\theta) \sum_{i=1}^n t_j(X_i) \right),$$

which factors into a function of the data alone and a function of $(T(\mathcal{X}), \theta)$. Sufficiency follows from the Factorization Theorem. $\square$

Notice what this theorem delivers: no matter how large the sample, the data can be compressed into a statistic whose dimension is k , fixed by the family rather than the sample size. This is the sense in which exponential families are the natural home of sufficiency. Not every family qualifies: the *Uniform* $[0, \theta]$ distribution cannot be written in exponential-family form because its support depends on θ — the same feature that makes it a recurring counterexample in this course. Exponential families will reappear: they are exactly the families with natural conjugate priors in Bayesian analysis (Lecture Note 12).

2.3 Minimal Sufficient Statistics

Could we reduce sufficient statistic $T(X)$ in the previous example even more? Suppose we have two statistics, say, $T(X)$ and $T^*(X)$. We say that T^* is not bigger than T if there exists some function r such that $T^*(X) = r(T(X))$. In other words, we can calculate $T^*(X)$ whenever we know $T(X)$. In this case when T^* changes its value, statistic T must change its value as well. In this sense, T^* provides at least as much data reduction as T .

Definition 3. A sufficient statistic $T^*(X)$ is called *minimal* if for any sufficient statistic $T(X)$ there exists some function r such that $T^*(X) = r(T(X))$.

Thus, in some sense, the minimal sufficient statistic gives us the greatest data reduction without a loss of information about parameters. The following theorem gives a characterization of minimal sufficient statistics:

Theorem 4. Let $f(x|\theta)$ be the pdf of X and $T(X)$ be such that, for any x, y , the statement $\{f(x|\theta)/f(y|\theta) \text{ does not depend on } \theta\}$ is equivalent to the statement $\{T(x) = T(y)\}$. Then $T(X)$ is minimally sufficient.

Example 14. Let us now go back to the example with $X_1, \dots, X_n \sim U[\theta, 1+\theta]$. Ratio $f(x|\theta)/f(y|\theta)$ is independent of θ if and only if $x_{(1)} = y_{(1)}$ and $x_{(n)} = y_{(n)}$ which is the case if and only if $T(x) = T(y)$. Therefore $T(X) = (X_{(1)}, X_{(n)})$ is minimal sufficient.

Example 15. Let $X_1, \dots, X_n$ be a random sample from the Cauchy distribution with parameter θ , i.e. the distribution with the pdf $f(x|\theta) = 1/(\pi(1+(x-\theta)^2))$. Then $f(x_1, \dots, x_n|\theta) = 1/(\pi^n \prod_{i=1}^n (1+(x_i - \theta)^2))$. By the theorem above, $T(X) = (X_{(1)}, \dots, X_{(n)})$ is minimal sufficient.

Here are three more examples of finding minimal sufficient statistics; each exercises the characterization theorem in a different parametric family.

Example 16. Let $X_1, \dots, X_n$ be a random sample from the distribution with pdf $f(x|\theta) = 2x/\theta^2$ for $0 < x < \theta$, where $\theta > 0$. The joint density is

$$f(x|\theta) = \prod_{i=1}^n \frac{2x_i}{\theta^2} I\{0 < x_i < \theta\} = \left(2^n \prod_{i=1}^n x_i\right) I\{x_{(1)} > 0\} \cdot \frac{1}{\theta^{2n}} I\{x_{(n)} < \theta\}.$$

By the factorization theorem with $h(x) = (2^n \prod_{i=1}^n x_i) I\{x_{(1)} > 0\}$ and $g(t|\theta) = \theta^{-2n} I\{t < \theta\}$, the sample maximum $T(X) = X_{(n)}$ is sufficient. For minimality, note that the ratio

$$\frac{f(x|\theta)}{f(y|\theta)} = \frac{\prod_{i=1}^n x_i}{\prod_{i=1}^n y_i} \cdot \frac{I\{x_{(n)} < \theta\}}{I\{y_{(n)} < \theta\}}$$

is independent of θ if and only if $x_{(n)} = y_{(n)}$. By the theorem above, $T(X) = X_{(n)}$ is minimal sufficient.

Example 17. Let $X_1, \dots, X_n$ be a random sample from the extreme value (Gumbel) location family with pdf $f(x|\theta) = e^{-(x-\theta)} \exp\{-e^{-(x-\theta)}\}$ for $-\infty < x < \infty$ and $-\infty < \theta < \infty$. The joint density is

$$f(x|\theta) = \exp\left\{-\sum_{i=1}^n (x_i - \theta)\right\} \exp\left\{-\sum_{i=1}^n e^{-(x_i - \theta)}\right\} = e^{n\theta} e^{-\sum_{i=1}^n x_i} \exp\left\{-e^\theta \sum_{i=1}^n e^{-x_i}\right\}.$$

By the factorization theorem with $h(x) = e^{-\sum_{i=1}^n x_i}$, the statistic $T(X) = \sum_{i=1}^n e^{-X_i}$ is sufficient. For minimality, note that the ratio

$$\frac{f(x|\theta)}{f(y|\theta)} = e^{-\sum_{i=1}^n x_i + \sum_{i=1}^n y_i} \exp\left\{-e^\theta \left(\sum_{i=1}^n e^{-x_i} - \sum_{i=1}^n e^{-y_i}\right)\right\}$$

is independent of θ if and only if $\sum_{i=1}^n e^{-x_i} = \sum_{i=1}^n e^{-y_i}$. By the theorem above, $T(X) = \sum_{i=1}^n e^{-X_i}$ is minimal sufficient. Notice that here the minimal sufficient statistic is neither a sample average nor an order statistic, but a sum of a non-linear transformation of the data.

Example 18. Let $X_1, \dots, X_n$ be a random sample from $\text{Binomial}(2, \theta)$, i.e. the distribution with pmf $f(x|\theta) = \frac{2!}{x!(2-x)!} \theta^x (1-\theta)^{2-x}$ for $x \in \{0, 1, 2\}$ and $0 \leq \theta \leq 1$. The joint pmf is

$$f(x|\theta) = \left(\prod_{i=1}^n \frac{2!}{x_i!(2-x_i)!}\right) \theta^{\sum_{i=1}^n x_i} (1-\theta)^{2n - \sum_{i=1}^n x_i},$$

so by the factorization theorem $T(X) = \sum_{i=1}^n X_i$ is sufficient. For minimality, the ratio

$$\frac{f(x|\theta)}{f(y|\theta)} = \frac{\prod_{i=1}^n (2!/(x_i!(2-x_i)!))}{\prod_{i=1}^n (2!/(y_i!(2-y_i)!))} \left(\frac{\theta}{1-\theta} \right)^{\sum_{i=1}^n x_i - \sum_{i=1}^n y_i}$$

is independent of θ if and only if $\sum_{i=1}^n x_i = \sum_{i=1}^n y_i$. By the theorem above, $T(X) = \sum_{i=1}^n X_i$ is minimal sufficient.

Remark 1. The idea behind sufficiency — reduce the data without losing the information relevant for the task at hand — has a famous analogue in causal inference. Suppose treatment $D_i \in \{0, 1\}$ is not randomly assigned, but its distribution depends on observed characteristics X_i through the *propensity score* $e(x) = P(D_i = 1 | X_i = x)$. Rosenbaum and Rubin (1983) show that the propensity score is a *balancing score*: conditional on $e(X_i)$, treatment is independent of the full covariate vector, $D_i \perp X_i | e(X_i)$ — and moreover it is the *coarsest* function of X_i with this property, just as a minimal sufficient statistic is the coarsest reduction of the data that remains sufficient.

2.4 Connection between Sufficient Statistics and MSE

Let $X = (X_1, \dots, X_n)$ be a random sample from distribution f_θ . Let $\hat{\theta} = \delta(X)$ be an estimator of θ . Let $T(X)$ be a sufficient statistic for θ . As we have seen already, an MSE provides one way to compare the quality of different estimators. In particular, estimators with smaller MSE are said to be more efficient. On the other hand, once we know $T(X)$, we can discard X . How do these concepts relate to each other?

The theorem below shows that for any estimator $\hat{\theta} = \delta(X)$, there is another estimator which depends on data X only through $T(X)$ and is at least as efficient as $\hat{\theta}$:

Theorem 5 (Rao-Blackwell). *In the setting above, define $\phi(T) = \mathbb{E}[\delta(X)|T]$. Then, $\hat{\theta}_2 = \phi(T(X))$ is an estimator for θ and $MSE(\hat{\theta}_2) \leq MSE(\hat{\theta})$. In addition, if $\hat{\theta}$ is unbiased, then $\hat{\theta}_2$ is unbiased and $V(\hat{\theta}_2) \leq V(\hat{\theta})$.*

Proof. To show that $\hat{\theta}_2$ is an estimator, we have to check that it does not depend on θ . Indeed, since T is sufficient for θ , the conditional distribution of X given T is independent of θ . So the conditional distribution of $\delta(X)$ given T is independent of θ as well. In particular, the conditional expectation $\mathbb{E}[\delta(X)|T]$ does not depend on θ . Thus, $\phi(T(X))$ depends only on the data X and $\hat{\theta}_2$ is an estimator.

We next establish the claim about MSE:

$$\begin{aligned} MSE(\hat{\theta}) &= \mathbb{E}[(\hat{\theta} - \hat{\theta}_2 + \hat{\theta}_2 - \theta)^2] \\ &= \mathbb{E}[(\hat{\theta} - \hat{\theta}_2)^2] + 2\mathbb{E}[(\hat{\theta} - \hat{\theta}_2)(\hat{\theta}_2 - \theta)] + \mathbb{E}[(\hat{\theta}_2 - \theta)^2] \\ &= \mathbb{E}[(\hat{\theta} - \hat{\theta}_2)^2] + 2\mathbb{E}[(\hat{\theta} - \hat{\theta}_2)(\hat{\theta}_2 - \theta)] + MSE(\hat{\theta}_2) \\ &= \mathbb{E}[(\hat{\theta} - \hat{\theta}_2)^2] + MSE(\hat{\theta}_2), \end{aligned}$$

where in the last line we used

$$\begin{aligned}
\mathbb{E}[(\hat{\theta} - \hat{\theta}_2)(\hat{\theta}_2 - \theta)] &= \mathbb{E}[(\delta(X) - \phi(T(X)))(\phi(T(X)) - \theta)] \\
&= \mathbb{E}[\mathbb{E}[(\delta(X) - \phi(T(X)))(\phi(T(X)) - \theta)|T]] \\
&= \mathbb{E}[(\phi(T(X)) - \theta)\mathbb{E}[(\delta(X) - \phi(T(X))|T]] \\
&= \mathbb{E}[(\phi(T(X)) - \theta) \cdot (\mathbb{E}[\delta(X)|T] - \phi(T(X)))] \\
&= 0,
\end{aligned}$$

since $\mathbb{E}[\delta(X)|T] = \phi(T(X))$.

To show the last result, we have

$$\mathbb{E}[\phi(T(X))] = \mathbb{E}[\mathbb{E}[\delta(X)|T]] = \mathbb{E}[\delta(X)] = \theta$$

by the law of iterated expectations. The claim about the variance follows from our bias-variance decomposition of MSE. $\square$

Example 19. Let $X_1, \dots, X_n$ be a random sample from $\text{Binomial}(k, p)$, i.e. $P\{X_j = m\} = (k!/(m!(k-m)!))p^m(1-p)^{k-m}$ for any integer $0 \leq m \leq k$. Suppose our parameter of interest is the probability of one success, i.e. $\theta = P\{X_j = 1\} = kp(1-p)^{k-1}$. One possible estimator is $\hat{\theta} = \sum_{i=1}^n I(X_i = 1)/n$. This estimator is unbiased, i.e. $\mathbb{E}[\hat{\theta}] = \theta$. Let us find a sufficient statistic. The joint density of the data is

$$\begin{aligned}
f(x_1, \dots, x_n) &= \prod_{i=1}^n (k!/(x_i!(k-x_i)!))p^{x_i}(1-p)^{k-x_i} \\
&= \text{function}(x_1, \dots, x_n)p^{\sum x_i}(1-p)^{nk-\sum x_i}.
\end{aligned}$$

Thus, $T = \sum_{i=1}^n X_i$ is sufficient. In fact, it is minimal sufficient.

Using the Rao-Blackwell theorem, we can improve $\hat{\theta}$ by considering its conditional expectation

given T . Let $\phi = \mathbb{E}[\hat{\theta}|T]$ denote this estimator. Then, for any nonnegative integer t ,

$$\begin{aligned}
\phi(t) &= \mathbb{E}\left[\sum_{i=1}^n I(X_i = 1)/n \mid \sum_{i=1}^n X_i = t\right] \\
&= \sum_{i=1}^n P\{X_i = 1 \mid \sum_{j=1}^n X_j = t\}/n \\
&= P\{X_1 = 1 \mid \sum_{j=1}^n X_j = t\} \\
&= \frac{P\{X_1 = 1, \sum_{j=1}^n X_j = t\}}{P\{\sum_{j=1}^n X_j = t\}} \\
&= \frac{P\{X_1 = 1, \sum_{j=2}^n X_j = t-1\}}{P\{\sum_{j=1}^n X_j = t\}} \\
&= \frac{P\{X_1 = 1\}P\{\sum_{j=2}^n X_j = t-1\}}{P\{\sum_{j=1}^n X_j = t\}} \\
&= \frac{kp(1-p)^{k-1} \cdot (k(n-1))!/((t-1)!(k(n-1)-(t-1))!)p^{t-1}(1-p)^{k(n-1)-(t-1)}}{(kn)!/(t!(kn-t)!)p^t(1-p)^{kn-t}} \\
&= \frac{k(k(n-1))!/((t-1)!(k(n-1)-(t-1))!)}{(kn)!/(t!(kn-t)!)} \\
&= \frac{k(k(n-1))!(kn-t)!t}{(kn)!(kn-k+1-t)!}
\end{aligned}$$

where we used the fact that X_1 is independent of $(X_2, \dots, X_n)$, $\sum_{i=1}^n X_i \sim \text{Binomial}(kn, p)$, and $\sum_{i=2}^n X_i \sim \text{Binomial}(k(n-1), p)$. So our new estimator is

$$\hat{\theta}_2 = \phi(X_1, \dots, X_n) = \frac{k(k(n-1))!(kn - \sum_{i=1}^n X_i)!(\sum_{i=1}^n X_i)}{(kn)!(kn - k + 1 - \sum_{i=1}^n X_i)!}.$$

By the theorem above, it is unbiased and at least as efficient as $\hat{\theta}$. The procedure we just applied is sometimes informally referred to as *Rao-Blackwellization*.

Note on Implementation: One may say “this is too complicated”. We have derived $\phi(t) = \mathbb{E}[\hat{\theta} \mid \sum_{i=1}^n X_i = t]$ analytically in order to calculate a new estimate $\hat{\theta}_2 = \phi(T) = \phi(\sum_{i=1}^n X_i)$, but in real life you may just do this with Monte-Carlo simulations. Note, that we do not need to calculate the whole $\phi(t)$ function we need only $\phi(T)$, that is evaluated in one point (the realized value of T). Note also, that the result does not depend on p , so we are free to choose any p (choosing p close to $T/(kn)$ will give faster calculations).

Choose some $p \in (0, 1)$. For $b = 1, \dots, B$ repeat the following:

- Draw $X_{1b}^*, \dots, X_{nb}^*$ as independent variables from $\text{Binomial}(k, p)$, if $\sum_{i=1}^n X_{ib}^* \neq T$, discard this sample. Repeat drawing samples until you get $\sum_{i=1}^n X_{ib}^* = T$.
- Calculate $Y_b = \frac{1}{n} \sum_{i=1}^n I(X_{ib}^* = 1)$.

The new estimator is $\hat{\theta}_2 \approx \frac{1}{B} \sum_{b=1}^B Y_b$. The accuracy is better for a larger number of simulations B .